



Supervised ML Workshop

DIT Pimpri



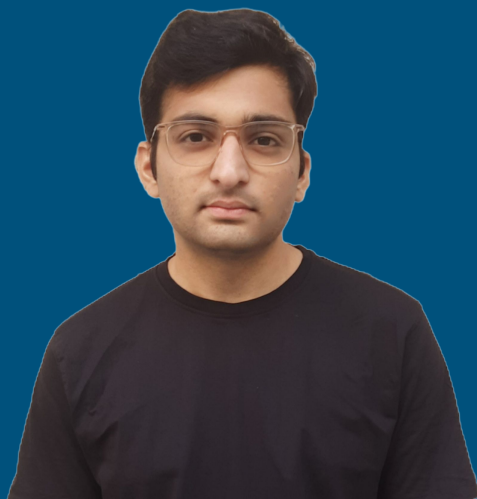
A workshop by Atharv, Ayush, Yash



Agenda for Workshop

Supervised Machine Learning

1. Overview of ML
 2. Linear Regression
 3. Logistic Regression
 4. Decision Trees 
 5. Ensemble Methods – Part 1:
 - Cascading
 - Stacking
 6. Ensemble Methods – Part 2:
 - Bagging
 - Boosting
-



Atharv

AI/ML Developer

GDG DIT - Machine Learning Lead

Works with the AI team at
ElevateTrust.AI

Always learning, building, and
sharing

Passionate about AI, ML, Generative
AI, EdgeAI



Ayush

AI/ML Developer

GDG DIT - Machine Learning Co-Lead

Turning data into insights with ML & AI, one model at a time.

Building apps where Machine Learning meets real-world impact.

Speedcuber — learned algorithms from cubes, patience (and suffering) from code.



Yash

AI/ML Developer

GDG-DIT Research Lead

Ex CP Member Lead

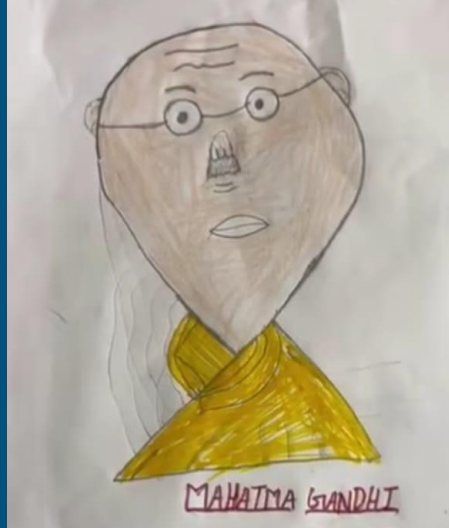
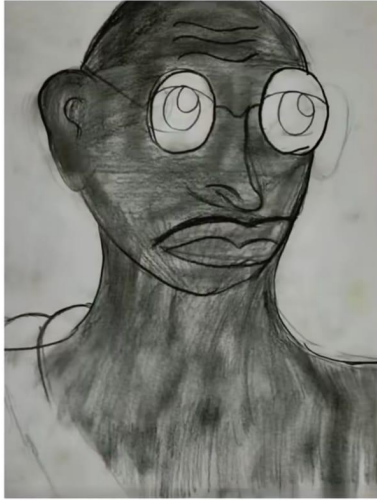
Ex Intern Accenture

Passionate about innovative tech

Forever Student







DECISION TREE

A decision tree is a model that uses a tree-like structure of questions to arrive at a decision.

A decision tree is basically a series of questions that split data into smaller, more similar groups.

Each question is a node, each answer is a branch, and the final outcome is a leaf.

TERMINOLOGIES

Root Node – The first node that represents the main condition.

Decision Node – A node where a condition is checked (like an if).

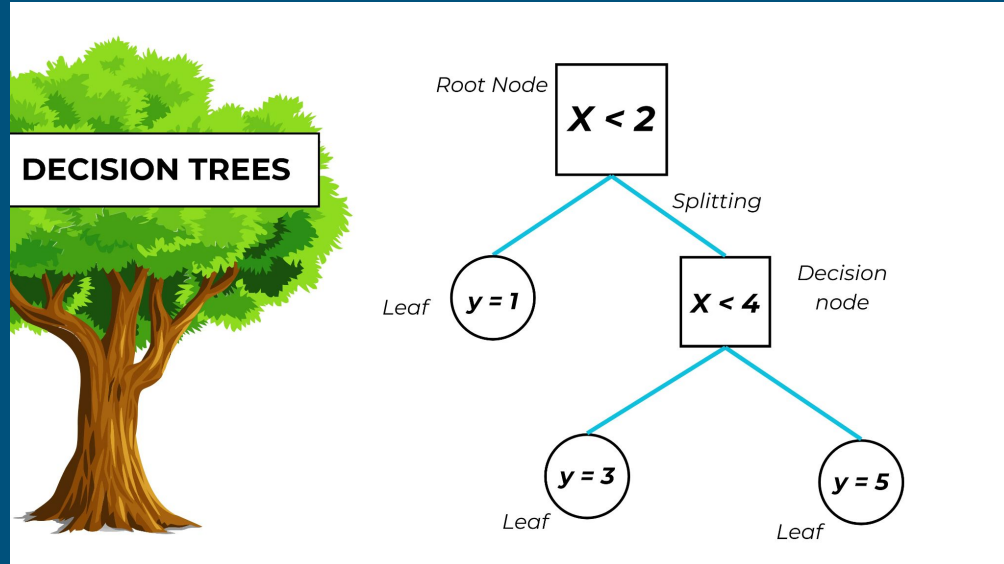
Leaf Node – The final outcome or decision (no further split).

Branch (Edge) – The path showing the result of a condition (if/else).

Splitting – Dividing a node into sub-nodes based on a condition.

Parent Node – A node that has child nodes.

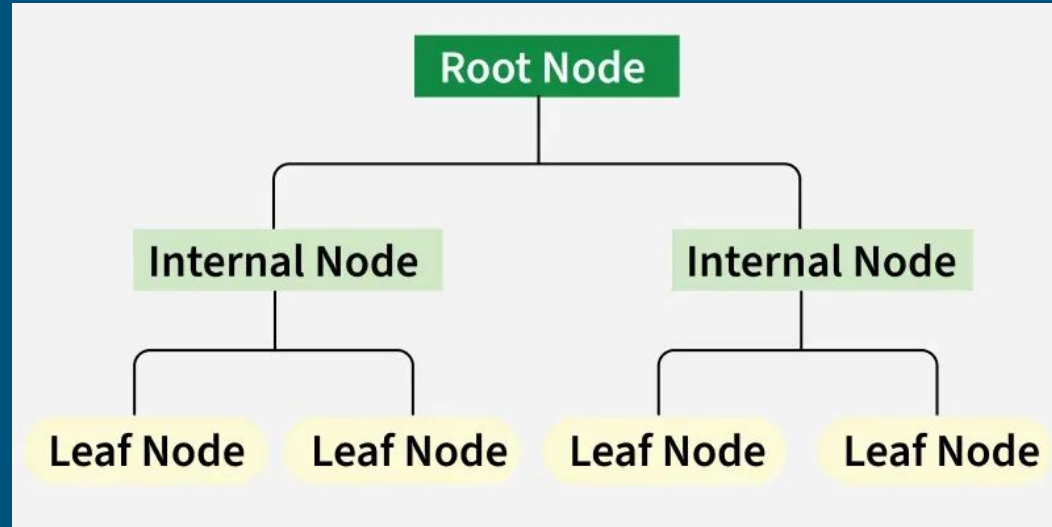
Child Node – A node that comes from a parent node after splitting.



Example 1: Going Out

```
if weather == "Sunny":  
    if have_money == True:  
        decision = "Go to Beach"  
    else:  
        decision = "Go for a Walk"  
else:  
    decision = "Stay Home"
```

Example 2:
Should i go to class?



Example 1

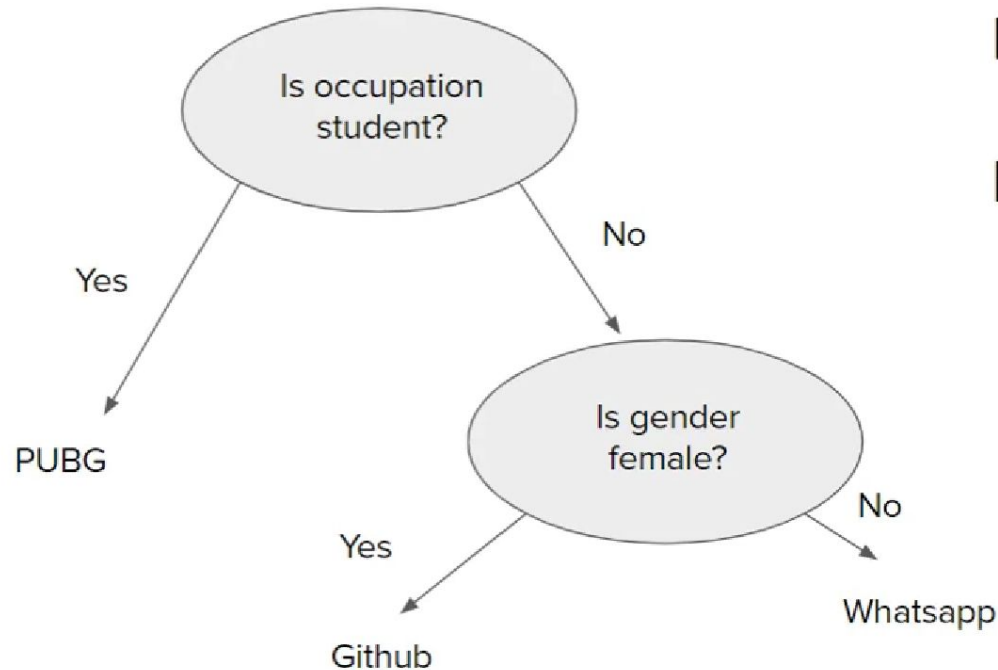
Gender	Occupation	Suggestion
F	Student	PUBG
F	Programmer	Github
M	Programmer	Whatsapp
F	Programmer	Github
M	Student	PUBG
M	Student	PUBG

Example 1

Gender	Occupation	Suggestion
F	Student	PUBG
F	Programmer	Github
M	Programmer	Whatsapp
F	Programmer	Github
M	Student	PUBG
M	Student	PUBG

```
If occupation==student
    print(PUBG)
Else
    If gender==female
        print(Github)
    Else
        print(Whatsapp)
```

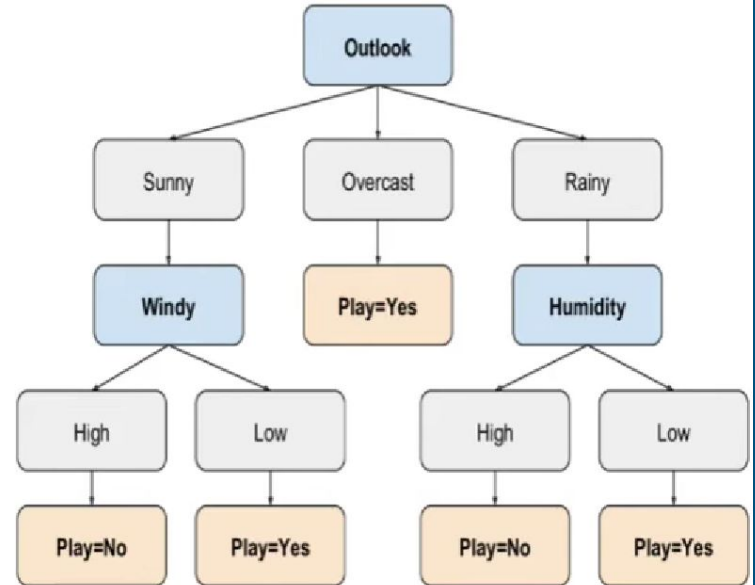
Where is the Tree?



```
If occupation==student
    print(PUBG)
Else
    If gender==female
        print(Github)
    Else
        print(Whatsapp)
```

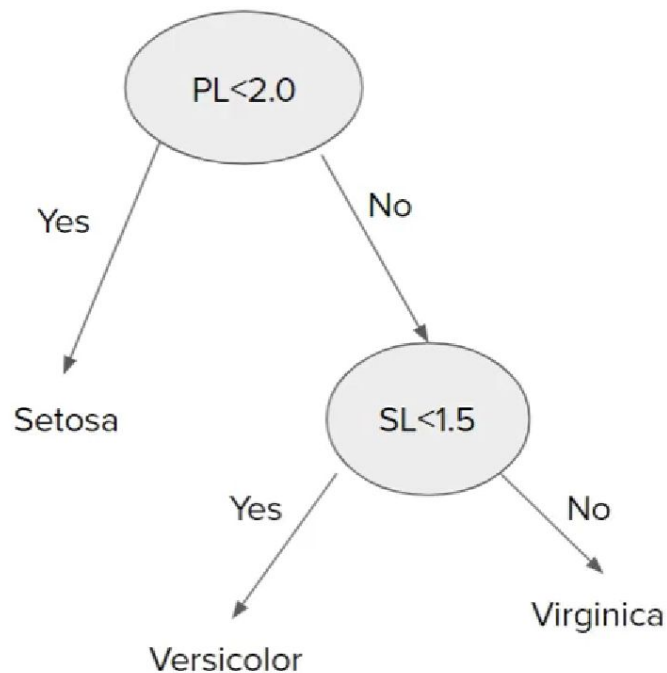
Example 2

Day	Outlook	Temp	Humid	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

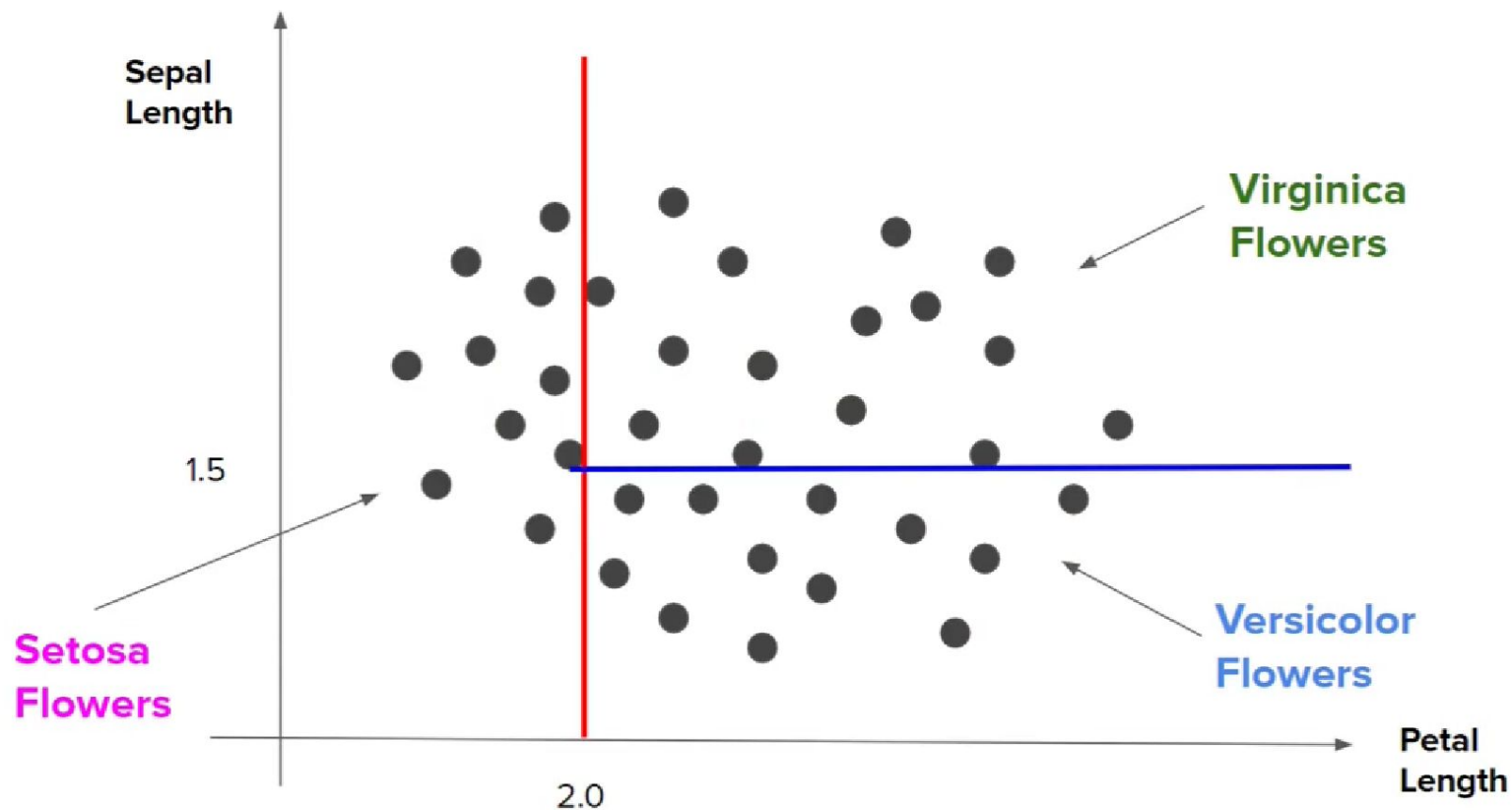


What if we have numerical data?

Petal Length	Sepal Length	Type
1.34	0.34	Setosa
3.45	1.45	Versicolor
1.69	0.98	Setosa
2.56	1.79	Virginica
3.00	1.13	Versicolor
1.3	0.88	Setosa



Geometric Intuition



What is a question ?



How good is a question ?

Ideal question should divide dataset into two perfect subset where each subset will have same label of their own.



Entropy

Entropy hame batata hai ki hamare dataset ka distribution kaisa hai
It gives us what is the uncertainty in the dataset
Ideally we want to reduce the uncertainty in the dataset

$$\begin{aligned}\text{Entropy} (\{y_1, \dots, y_n\}) &= \text{Entropy} (p) \\ &= -p \log(p) + (1-p) \log(1-p)\end{aligned}$$

Information Gain

$$\text{Entropy}(D) - [\text{Gamma} * \text{Entropy}(D_{\text{yes}}) + (1 - \text{Gamma}) * \text{Entropy}(D_{\text{no}})]$$

$$\text{Gamma} = \frac{|D_{\text{yes}}|}{|D|}$$

Algorithm

1. Discretize each feature in $[\min, \max]$ range
2. Pick question that has largest information gain
3. Repeat the procedure for Dyes, Dno
4. Stop building a tree if node becomes SUFFICIENTLY pure

Alternate measures for goodness of a question - GINI INDEX

Assignment

1. Explore more on Linear Regression refer this video
<https://youtu.be/ilkJrwVUI1c>
2. Explore Polynomial Regression

Connect with us on LinkedIn



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Machine Learning · Deep Learnin...



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Ex-Intern @ Accenture | AI & DS
Student



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Thank You